

ET AI HACKATHON 2026 · PROBLEM STATEMENT 8

ForgeMind AI

Unified Asset & Operations Brain for Industrial Knowledge

Plant docs are scattered across manuals, inspections, work orders, and incident reports.
ForgeMind puts them in one place so teams can upload, search, and ask — with sources.

Industrial knowledge is fragmented. ForgeMind unifies it.

THE PROBLEM

35% of engineer time lost searching for information that already exists (McKinsey, 2024).

7–12 disconnected document systems per large industrial plant.

18–22% of unplanned downtime traced to fragmented equipment history.

25% of India's experienced engineers retire within a decade — taking undocumented knowledge with them.

THE SOLUTION

Upload — PDF/Word/Excel/text, parsed and persisted for the whole team.

Knowledge — keyword + semantic search, equipment map, full asset history.

Ask AI — plain-language Q&A with cited sources and grounded compliance.

Insights — live coverage score, auto-detected conflicts/gaps, Lessons Learned.

A live intelligence layer, not a static demo

1 Upload → parse text → LLM entity extraction (equipment tags, parameters, regs, personnel, dates)

2 SQLite (documents + entities) + vector index (local embeddings, optional ChromaDB)

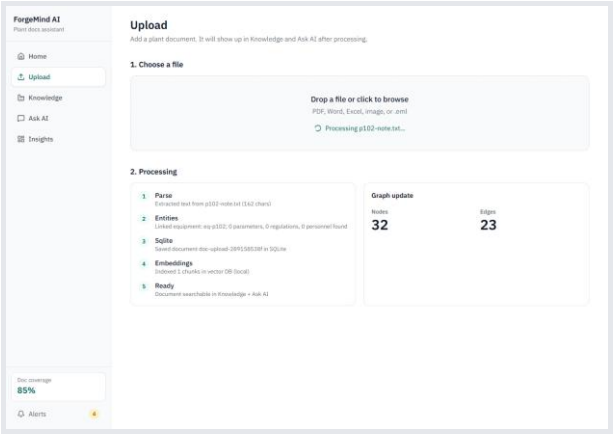
3 Knowledge graph grows · conflicts auto-detected · gaps recomputed — live, per upload

4 Ask AI (RAG) → OpenRouter LLM → answer grounded in retrieved regulation/SOP evidence

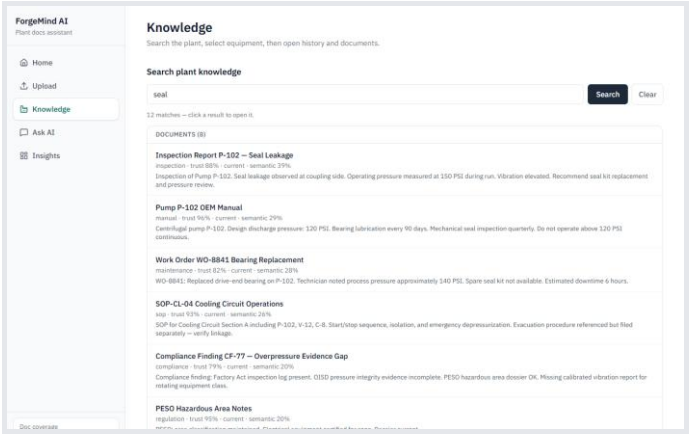
WHY IT'S NOT JUST A DEMO

- Uploading a real document adds real graph nodes/edges, detects real value conflicts, and closes real doc-type gaps — verified end-to-end.
- Compliance flags are grounded in retrieved evidence and marked 'unknown' rather than guessed when none exists.

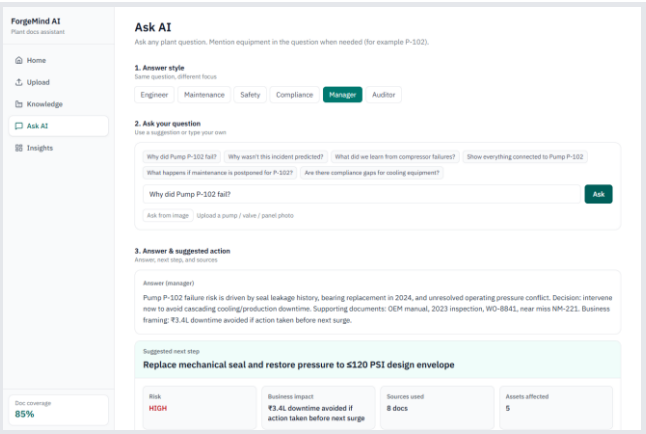
Upload → Knowledge → Ask AI → Insights



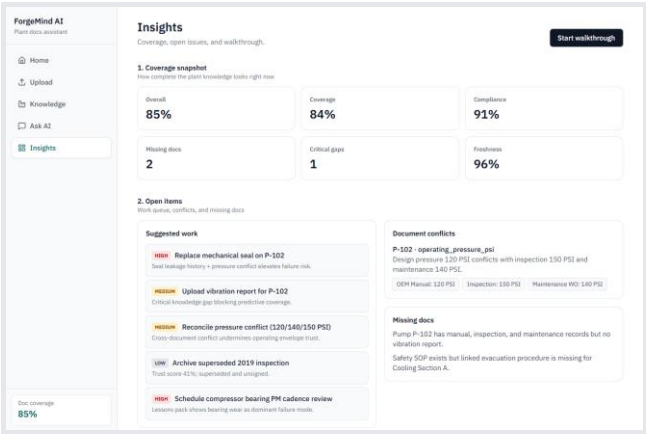
Upload — live processing stages, graph grows with every file



Knowledge — keyword + semantic search across the plant



Ask AI — grounded answer with a decision card and sources



Insights — coverage, conflicts, gaps, and lessons learned

Mapped directly to what was asked

Innovation — 25%	Live-growing knowledge graph, automatic conflict/gap detection.
Business Impact — 25%	Reclaims the 35% of time lost to document search; faster RCA & compliance evidence.
Technical Excellence — 20%	Real RAG, grounded compliance, resilient error handling (no hallucinated pass/fail, no CORS-masked 500s).
Scalability — 15%	SQLite → Postgres/Qdrant swap-in scaffolded; OpenRouter is model-agnostic.
User Experience — 15%	Guided walkthrough, real graph visualization, tested end-to-end with zero console errors.

Thank you

Try it live:

<https://forgemind.tarachand.tech>

github.com/tarungarg18/ForgeMind-AI

Next: OCR for scanned P&IDs, async ingestion at scale, knowledge-graph-native storage, role-based access, and multi-plant federation.